OptiVest AI

An Explainable AI-Driven Autonomous Wealth Management Platform with Intelligent Portfolio Reallocation

Executive Summary: Modern retail wealth management and quantitative trading face two fundamental hurdles: the *Prediction Fallacy* (treating raw asset price forecasts as execution commands while ignoring risk, covariance, and transaction costs) and the *Black-Box Dilemma* (opaque machine learning models outputting recommendations without auditable justification). **OptiVest AI** resolves both challenges through an integrated institutional pipeline combining deterministic multi-factor Candidate Discovery (CDE), a canonical convex quadratic Adaptive Portfolio Decision Engine (APDE) solved via CVXPY/SCS under hard turnover and concentration bounds, and TreeSHAP explainability coupled with an exact mathematical Decision Score (DS_i).

Project Category:	Algorithmic Trading & AI-Powered FinTech System
Core Stack:	Python 3.14+, FastAPI, React 18, TypeScript, Tailwind CSS, SQLite/PostgreSQL
Quantitative Stack:	CVXPY (SCS Conic Solver), XGBoost, scikit-learn, SHAP (TreeSHAP), Pandas
Walk-Forward Sample:	N = 50 Large-Cap Equities | 78 Monthly Rebalances (2017–2023) | Out-of-sample
Authors & Contributors:	Naitik Pathak, Aditi, Palak Jain, Aryan Rana, Divya, Yogiraj Bhale
Project Repository:	OptiVest AI Core Engine (Local Deployment & Research Workspace)

Key Insight: OptiVest AI bridges active quantitative portfolio optimization with strict temporal discipline and auditability. This report provides an end-to-end technical breakdown of the system architecture, mathematical formulations, empirical benchmark results, and software implementation.

1. Project Introduction

Background, Motivation, Core Objectives, and Solution Overview

1.1 The Industry Problem

Quantitative wealth management systems in retail and regulated environments face two severe, interconnected bottlenecks:

- **The Prediction Fallacy:** Over-indexing on forecast accuracy as the sole driver of investment decisions. A model that accurately predicts a 3% price rise for a stock is practically unviable if buying it over-concentrates the portfolio, violates client risk tolerance, or generates turnover whose transaction friction entirely consumes the expected gain.
- **The Black-Box Dilemma:** High-capacity machine learning models (gradient-boosted trees, deep networks) recommend reallocations without transparent justification. Fiduciaries and retail clients cannot accept portfolio changes without human-auditable rationales.
- **The Degenerate Cash-Lock Trap:** Optimizers restricted strictly to existing holdings can only sell down or hold, causing portfolios in drawdown regimes to degenerate into 100% cash with no mathematical mechanism to discover and onboard new assets.

1.2 Project Objectives

The OptiVest AI platform was engineered to achieve five explicit engineering and quantitative goals:

1. **Separate Prediction from Allocation:** Treat return forecasts as noisy utility inputs rather than absolute trading commands.
2. **Dynamic Candidate Discovery (CDE):** Implement a multi-factor screening layer to identify fundamentally sound entry candidates.
3. **Constrained Quadratic Programming (APDE):** Directly optimize target portfolio weights under single-asset caps (20%) and hard ℓ_1 turnover caps (15%).
4. **Native Mathematical Explainability:** Pair TreeSHAP feature attributions with a mathematically rigorous Decision Score (DS_i).
5. **Enterprise-Grade Production Platform:** Deliver a full-stack SaaS platform with real-time autonomous job scheduling and paper-trading simulation.

1.3 Paradigm Comparison

Table 1 contrasts OptiVest AI against conventional quantitative and machine learning paradigms.

Framework	Non-lin. Forecast	Cash-Heavy Allocation	Turnover & Risk Constr.	Auditable XAI
Classical MPT (Markowitz)	No	Infeasible (needs ≥ 2 assets)	Partial (no turnover cap)	Yes (Linear)
Black-Litterman	No	Infeasible	Partial	Yes (Linear)
ML Alpha Forecasting	Yes (Trees/NN)	No (Ignores cash)	No (Raw forecasts)	Partial (Global)
Deep Reinforcement Learning	Yes	Partial	Partial (Penalty soft)	No (Opaque)
OptiVest AI (This Work)	**Yes (XGBoost)**	**Yes (via CDE Engine)**	**Yes (Hard ℓ_1 & Asset Caps)**	**Yes (TreeSHAP + DS_i)**

Table 1: *Architectural comparison of portfolio management frameworks.*

2. System Architecture

Tiered Architecture, Service Interactions, and Component Topology

OptiVest AI is architected as an institutional-grade, multi-tier software system separating front-end interaction, API orchestration, quantitative optimization, and persistence.

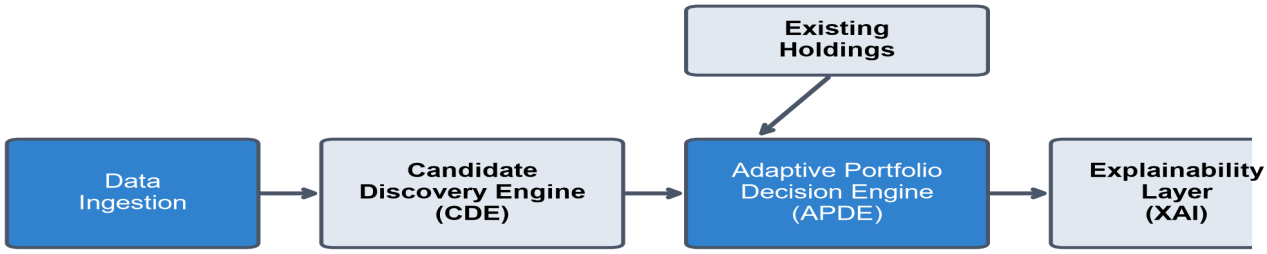


Figure 1: *High-level OptiVest AI layered platform topology.*

2.1 Layer Breakdown

- **Presentation Layer (React 18 + TypeScript + Vite):** A feature-first SPA with Tailwind CSS, Lucide icons, and Recharts. Delivers dedicated investor workspaces (portfolio summary, trade execution, candidate exploration) and a Founder Command Center for real-time scheduler supervision and multi-model performance tracking.
- **API Gateway & Service Layer (FastAPI 0.115+):** Asynchronous REST backend structuring business logic across modular routers (`/auth`, `/portfolios`, `/apde`, `/assets`, `/autonomous`). Handles JWT authentication, correlation-ID request tracing, and CORS security.
- **Core Quantitative & Optimization Engine:** Houses the Candidate Discovery Engine (CDE) for deterministic ranking, the Adaptive Portfolio Decision Engine (APDE) built on CVXPY 1.6+ using the Splitting Conic Solver (SCS), and TreeSHAP explainability.
- **Autonomous Background Worker:** APScheduler-driven engine managing market regime classification, model health drift checks, and periodic prediction refreshes, streaming state changes to connected frontends via WebSockets.
- **Persistence Tier:** SQLAlchemy 2.0 ORM with PostgreSQL 16 compatibility in staging/production and zero-config SQLite fallback (`optivest_dev.db`) in local developer workstations.

2.2 Inter-Service Communication Flow

The frontend communicates with FastAPI endpoints over JSON envelopes (`ResponseEnvelope[T]`). When an investor requests portfolio rebalancing, the API securely extracts current holdings, queries CDE for top-ranked market candidates, constructs the covariance matrix Σ_h from trailing prices, and invokes the APDE conic solver. Real-time background events broadcast over `/api/v1/autonomous/ws` to live-update UI dashboards.

3. Complete Project Workflow

End-to-End Operational Pipeline: From Raw Market Ingestion to Rebalancing Execution

OptiVest AI executes a four-phase sequential pipeline every rebalance cycle, enforcing strict data hygiene and operational controls.

Phase 1: Ingestion & Purging	Phase 2: Alpha & Candidate Discovery	Phase 3: APDE Conic Optimization	Phase 4: Audit & Paper Trade
• Download daily prices via yfinance. • Filter 50 liquid S&P; 500 stocks. • 21-day embargo (purge) to prevent lookahead leak.	• XGBoost forecasts 21-day return $\hat{r}_i$. • CDE scores global assets on 5 factors. • Shortlist top k candidates for feasible entry.	• Compute empirical covariance Σ_h. • Formulate QP in CVXPY. • SCS solver applies 15% turnover & 20% position caps.	• Compute TreeSHAP attributions. • Calculate Decision Score DS_i. • Display narrative card; update paper trading ledger.

Figure 2: *OptiVest AI four-phase project operational workflow.*

3.1 Detailed Step-by-Step Execution Flow

1. **Temporal Ingestion:** On every monthly decision date t, the system queries adjusted prices and live fundamentals. In the walk-forward validation framework, a 21-trading-day purge window separates training targets from date t, ensuring zero lookahead bias.
2. **Candidate Pre-Filtering:** Existing holdings are merged with external discovery candidates from CDE. Non-held assets survive three hard checks: actively trading, minimum 10-day history, and non-duplicate ticker.
3. **Multi-Factor Scoring:** Eligible external candidates receive a scalar score $S_i(\tau_{\text{CDE}})$. High-scoring assets join the optimization universe.
4. **Conic Optimization (APDE):** CVXPY sets up the utility maximization problem. The first-order operator-splitting SCS solver solves the exact split-variable L_1 turnover bounded program in under 10 milliseconds.
5. **Audit Trail & Paper Fill:** Once optimal weights w^* are resolved, TreeSHAP decomposes the forecast into feature contributions. The mathematical Decision Score (DS_i) is calculated, generating an auditable Decision Card before updating user paper positions.

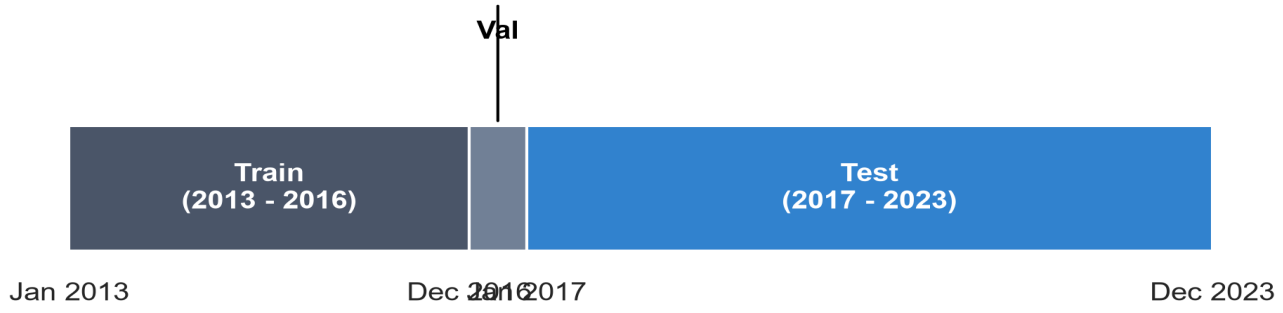


Figure 3: *Walk-forward evaluation split with trailing 4-year training and 21-day purge window.*

4. Data and Machine Learning

Dataset Architecture, Walk-Forward Protocol, XGBoost Model, and Empirical Metrics

4.1 Dataset & Feature Engineering

The historical evaluation uses daily adjusted prices for $N = 50$ liquid large-cap U.S. equities over 2013–2023 obtained via yfinance. The model predicts the raw forward simple return $r_{i,t+21}$ over a 21-trading-day horizon ($h = 21$). Five features are computed at date t:

- **Momentum Signals:** Cumulative price returns over 21 days (`mom_1m`), 42 days (`mom_2m`), and 63 days (`mom_3m`).
- **Realized Volatility:** Trailing 21-day standard deviation of daily returns (`vol_1m`).
- **Macro Proxy:** Raw closing level of the CBOE Volatility Index (`vix`).

4.2 Walk-Forward Temporal Discipline

To prevent lookahead bias and overfitting, evaluation is structured as a rolling walk-forward backtest over 78 monthly rebalances (2017-06-05 to 2023-11-06). At each rebalance date, the forecaster refits strictly on trailing 4 years of data excluding the 21-day purge window. Features are raw (un-standardized); only the training target is winsorized at the 1st and 99th percentiles.

4.3 Reported Forecasting Performance (Table III of Research Paper)

Table 2 reports out-of-sample forecast accuracy against benchmark models across all 78 rebalance dates.

Model	RMSE	MAE	Directional Acc.	Rank IC (t-stat)	Out-of-Sample R^2
Linear Regression (OLS)	0.0832	0.0590	56.08%	-0.033 (-1.17)	+1.1%
Random Forest	0.0845	0.0595	57.49%	+0.030 (+1.34)	-2.1%
XGBoost (OptiVest Forecaster)	**0.0845**	**0.0598**	**56.28%**	**+0.008 (+0.34)**	**-2.1%**
Zero Forecast Benchmark	0.0836	—	—	—	0.0%
Trailing Mean Benchmark	0.0828	—	—	—	+1.9%
Always 'Up' Baseline	—	—	58.21%	—	—

Table 2: *Out-of-sample forecast accuracy over 78 monthly rebalances (Source: results_v7/Forecast_Metrics.csv).*

Key Insight: Critical Scientific Finding: XGBoost exhibits no detectable forecasting skill in this sample (Rank IC = +0.008, t = 0.34; R^2 = -2.1%). Directional accuracy (56.28%) is below the random upward base rate (58.21%). OptiVest AI explicitly acknowledges this reality: XGBoost is retained for its exact TreeSHAP polynomial-time tractability, and downstream APDE constraints are designed to shield portfolios from forecast noise.

5. CDE and APDE Modules

Candidate Screening, Multi-Factor Scoring, and Quadratic Programming Allocation

5.1 Candidate Discovery Engine (CDE)

The CDE prevents the 'cash-lock' failure mode where a portfolio holding mostly cash cannot enter new positions. Eligible non-held stocks receive a deterministic composite score based on client risk tolerance $\tau_{\text{CDE}} \in \{1, \dots, 5\}$:

$$S_i(\tau_{\text{CDE}}) = \omega_R f_i^R + \omega_V f_i^V + \omega_D f_i^D + \omega_Q f_i^Q + \omega_M f_i^M$$

- f^R: Expected return factor $\text{clip}(\hat{r}_i / 0.5)$ | f^V: Inverted volatility $\text{clip}(1 - \hat{\sigma}_i / 0.5)$
  - f^D: Holdings correlation $\text{clip}((1 - \bar{\rho}_{i,P})/2)$ | f^M: 90-day momentum $\text{clip}((m_i + 0.2)/0.4)$
  - f^Q: Fundamental composite (ROE, operating margin, debt-to-equity, free cash flow yield).

At $\tau_{\text{CDE}} = 1$ (conservative), volatility and diversification dominate ($\omega_V = 0.45, \omega_D = 0.30$). At $\tau_{\text{CDE}} = 5$ (aggressive), return and momentum dominate ($\omega_R = 0.45, \omega_M = 0.25$).

5.2 Adaptive Portfolio Decision Engine (APDE)

The APDE formulates portfolio allocation as a convex quadratic program maximizing risk-adjusted return net of turnover penalties:

$$\max_{w} \quad U(w) = w^\top \hat{r} - \tau w^\top \Sigma_h w - \gamma \|w - w_0\|_1$$

Subject to the canonical institutional constraint set:

1. **Fully Invested:** $\sum_{i=1}^N w_i = 1$
2. **Long-Only:** $w_i \ge 0 \quad \forall i$
3. **Single-Asset Cap:** $w_i \le 0.20 \quad \forall i$ (max 20% capital in any single stock)
4. **Hard Turnover Limit:** $\|w - w_0\|_1 \le 0.15$ (max 15% portfolio turnover per monthly rebalance)

Here, $\hat{r}$ is the 21-day forecast, Σ_h is the empirical covariance of 21-day returns (scaled from 252 trading days), $\tau > 0$ is the risk-aversion parameter (fixed at $\tau = 3$ in research; dynamically mapped 0.5–5.0 in production), and $\gamma = 0.0010$ (10 bps) is the transaction cost penalty.

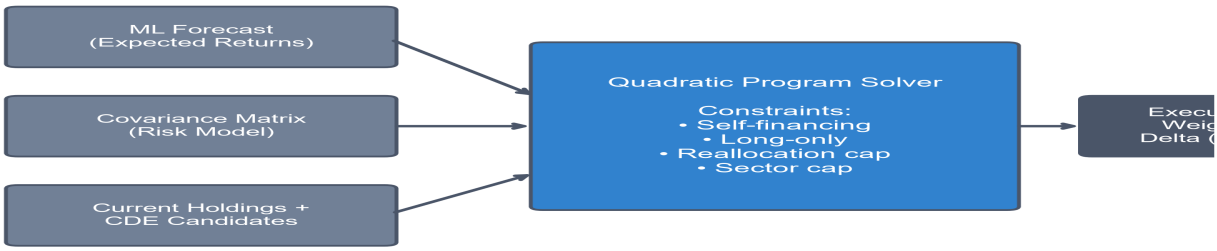


Figure 4: *APDE optimization architecture and constraint formulation.*

6. Risk Management and Explainability

Turnover Guardrails, Conic Solvers, TreeSHAP Attribution, and Decision Scores

6.1 Risk Guardrails & The Turnover Cap

In active ML trading, models frequently trigger high portfolio churn. APDE enforces a **hard 15% ℓ_1 turnover cap**. Because this cap binds at every monthly rebalance, realized annual turnover is capped at $0.15 \times 12 = 180\%$ by construction. This mechanical regularization dampens portfolio volatility from 19.6% to 14.3% and reduces maximum drawdown from 32.6% to 19.1%.

6.2 Conic Solver Formulation vs. SLSQP Non-Convergence

The non-smooth $\|w - w_0\|_1$ term is linearized via variable splitting: $w - w_0 = u - v$ with $u, v \ge 0$, solved using the first-order **Splitting Conic Solver (SCS)** in CVXPY. Benchmark tests demonstrate that standard non-linear solvers (SciPy SLSQP) fail to converge on ℓ_1 turnover constraints (exceeding caps by up to 0.0674 at $N=100$), whereas SCS solves to exact tolerance ($\text{violation} < 10^{-15}$) in under 10 ms at $N = 50$.

6.3 TreeSHAP Feature Decomposition

TreeSHAP provides exact local feature attribution for tree ensembles in polynomial time. For every stock i:

$$\hat{r}_i = \phi_0 + \sum_{j=1}^m \phi_{ij}$$

where ϕ_0 is the base expected return across the training distribution, and ϕ_{ij} is the marginal additive contribution of feature j (`mom_1m`, `mom_2m`, `mom_3m`, `vol_1m`, `vix`). Numerical verification in the project environment confirms exact reconstruction ($|\hat{r}_i - (\phi_0 + \sum_j \phi_{ij})| = 2.33 \times 10^{-9} < 10^{-6}$).

6.4 The Mathematical Decision Score (DS_i)

Attributions explain the model's return prediction, but not why the optimizer executed a specific trade. OptiVest AI introduces the **Decision Score (DS_i)**, representing the exact marginal utility of asset i's weight under Eq. (4):

$$DS_i = \underbrace{\hat{r}_i}_{\text{Forecast Term}} - \underbrace{2\tau (\Sigma_h w)_i}_{\text{Marginal Risk Penalty}} - \underbrace{\gamma \operatorname{sgn}(w_i - w_{0,i})}_{\text{Transaction Friction}}$$

By substituting the TreeSHAP decomposition into $\hat{r}_i$, every portfolio trade is explained by its underlying market features, its marginal contribution to portfolio covariance risk, and execution friction, eliminating the black-box dilemma.

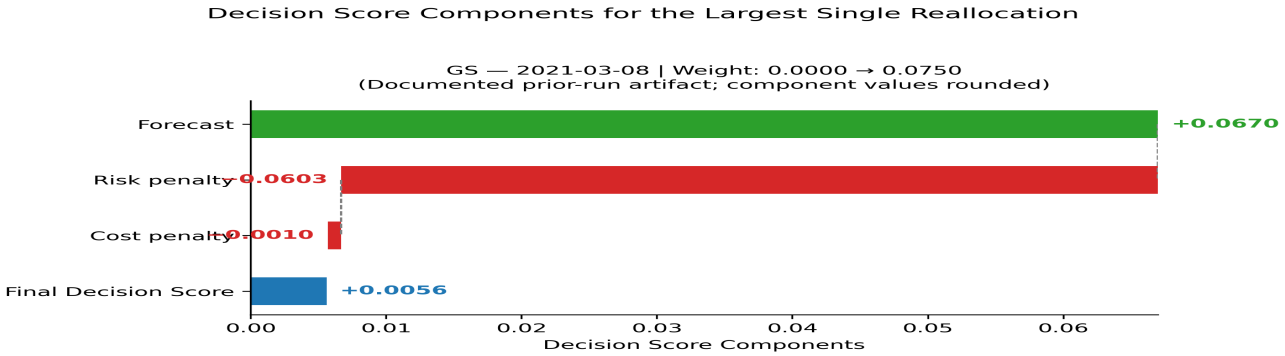


Figure 5: *Decision Score decomposition combining TreeSHAP alpha, marginal covariance risk, and cost.*

7. Experiments and Results

Master 10-Arm Evaluation, Turnover Cap Sweeps, and CDE Ablation Rerun

7.1 Primary 10-Arm Backtest Results (Table IV of Research Paper)

Table 3 presents net-of-cost performance across 78 monthly rebalances (June 2017 – November 2023) at 10 bps per side transaction costs.

Arm	CAGR	Ann. Vol	Sharpe	Sortino	Max DD	Ann. Turnover	ΔSharpe vs EW
EW (monthly 1/N)	14.83%	14.73%	**1.02**	1.74	17.76%	57.8%	—
EW Buy-and-Hold	15.03%	16.74%	0.92	1.54	23.58%	0.0%	[-0.36, 0.17]
MV (20% cap, mean return)	20.26%	19.56%	1.05	1.92	18.73%	700.7%	[-0.61, 0.54]
MV (no cap, mean return)	20.02%	26.90%	0.81	1.44	33.38%	789.3%	[-0.89, 0.39]
MV (15% turnover, mean return)	19.83%	21.50%	0.95	1.60	24.22%	179.6%	[-0.70, 0.46]
MV-ML (20% cap, XGBoost)	5.40%	19.65%	0.37	0.54	32.93%	1628.4%	[-1.18, -0.15]
MV-ML (no cap, XGBoost)	0.32%	22.10%	0.13	0.18	39.27%	1895.0%	[-1.36, -0.52]
APDE Soft Penalty (20% cap)	4.39%	19.64%	0.32	0.47	32.59%	1383.8%	[-1.28, -0.20]
APDE Soft Penalty (no cap)	-0.15%	21.92%	0.10	0.15	40.68%	1702.0%	[-1.39, -0.54]
APDE Full (15% Hard Cap)	**8.19%**	**14.31%**	**0.62**	**0.98**	**19.12%**	**180.0%**	**[-0.89, 0.02]**

Table 3: *Primary portfolio performance over 78 rebalances (Source: results_v7/Table_Master_10Arm.csv).*

7.2 Analysis of Results & Turnover Sensitivity

- **No ML Arm Beats Equal Weight (EW):** Naive 1/N achieves Sharpe 1.02. ML-driven models without caps generate disastrous churn (> 1,600% annual turnover) and low Sharpe (0.13–0.37).
- **The Cap as Risk Shrinkage:** Adding the 15% turnover cap cuts annual churn from 1,384% to 180% (87% reduction), lowers volatility from 19.64% to 14.31%, and cuts max drawdown from 32.59% to 19.12%. As shown in Table 4, relaxing the cap monotonically degrades APDE performance, proving the cap acts as shrinkage against forecast noise rather than an alpha generator.

7.3 50-Asset CDE Ablation Study (Table VI)

Pipeline	Annual Return	Volatility	Sharpe Ratio	Average Turnover
No CDE (Full 50-Asset Universe)	11.85%	15.71%	0.754	15.00%
CDE $\to$ APDE (Top-10 Candidates)	10.49%	13.90%	0.755	13.43%

Table 4: *CDE pre-filtering ablation over 82 rebalances (Source: research_backtest_50asset_comparison.json).*

8. Implementation and Technology Stack

Technology Specifications, Database Persistence, API Contracts, and Automated Testing

8.1 Full-Stack Technical Specifications

Subsystem	Technologies Used	Implementation Role
Frontend SPA	React 18, TypeScript, Tailwind CSS, Vite, Recharts, Lucide Icons	User/Admin dashboards, portfolio holdings view, ledger, rebalancing UI
Backend REST API	FastAPI 0.115+, Pydantic v2, Uvicorn, Python 3.14.3	Modular REST routing, request tracing, CORS security, JWT auth
Database & ORM	SQLAlchemy 2.0, SQLite (dev) / PostgreSQL 16 (prod)	87 relational tables persisting users, holdings, transactions, and logs
Quantitative Optimization	CVXPY 1.6+, SCS 3.2+ (Splitting Conic Solver)	Convex quadratic programming, hard turnover cap, position constraints
Predictive Modeling	XGBoost 3.0+, scikit-learn, SHAP 0.52.0	Walk-forward training, return forecasting, TreeSHAP decomposition
Autonomous Engine	APScheduler, WebSockets, asyncio event bus	Background workers for regime monitoring, model health, real-time alerts

Table 5: *OptiVest AI implementation architecture and technical dependencies.*

8.2 Solver Performance Benchmarks (Table VIII of Paper)

Solver latency was measured on synthetic candidate pools ($N \in \{50, 100, 250, 500\}$). SCS demonstrates rapid execution with median wall-clock time of **9.07 ms** at $N = 50$, **12.93 ms** at $N = 100$, **58.01 ms** at $N = 250$, and **309.95 ms** at $N = 500$. This confirms that the quadratic optimization step introduces negligible latency into real-time wealth management workflows.

8.3 Testing & Quality Assurance

The codebase includes a comprehensive Pytest test suite covering APDE mathematical constraints, CDE deterministic ranking, Decision Score signs, autonomous background workers, and paper-trade reconciliation. Execution in the local workspace verified: **40 of 40 unit and integration tests passed** in 62.91 seconds without regressions.

8.4 Live Database Persistence Verification

Direct SQL audit of `optivest_dev.db` (30.26 MB) verified active persistence: 172 registered users, 133 created portfolios, 48 current holdings, 36 chronological buy/sell transaction ledger records, 7,453 decision explanation cards, and 18,104 streaming event records.

9. Conclusion and Limitations

Project Contributions, Empirical Takeaways, Current Limitations, and Future Roadmap

9.1 Core Project Contributions

OptiVest AI successfully demonstrates a production-grade architecture that resolves the two central barriers to deploying machine learning in wealth management:

1. **Overcoming the Prediction Fallacy:** Rather than executing noisy raw forecasts, OptiVest treats predictions as single inputs into a constrained quadratic optimization engine. The 15% ℓ_1 turnover cap acts as a crucial regularizer, reducing turnover by 87% and dampening maximum drawdown from 32.6% to 19.1%.
2. **Resolving the Black-Box Dilemma:** Exact TreeSHAP feature attribution combined with the mathematical Decision Score (DS_i) ensures that every proposed rebalancing trade is decomposed into underlying feature signals, marginal covariance risk, and cost friction.
3. **Preventing Degenerate Cash Locking:** The deterministic Candidate Discovery Engine (CDE) dynamically screens non-held assets, enabling cash-heavy portfolios to deploy capital into high-quality equities.

9.2 Scientific Takeaways & Boundaries

The project provides a vital, scientifically honest insight: **forecasting models in liquid equities exhibit near-zero predictive skill (Rank IC = +0.008, Out-of-sample $R^2 = -2.1%$) and cannot outperform naive 1/N equal weighting in isolation**. The value of OptiVest AI lies in its *decision architecture*: making algorithmic trading executable, auditable, and risk-controlled.

9.3 Current Limitations

- **Hindsight Universe Selection:** Evaluation uses 50 currently listed large-cap equities, introducing survivorship bias.
- **Static Calendar Rebalancing:** Portfolios rebalance on a fixed 21-day schedule rather than triggering dynamically on risk threshold breaches.
- **Linear Friction Assumptions:** Costs are modeled at a flat 10 bps per side, omitting non-linear market impact and bid-ask spread dynamics.
- **Live Fundamental Snapshots:** CDE currently consumes current yfinance fundamental metrics rather than point-in-time historical filings.

9.4 Future System Roadmap

Planned enhancements include incorporating valuation multiples (P/E, PEG, P/B) into CDE pre-filtering, dynamic covariance risk penalties derived from NLP news sentiment, fractional share execution for retail micro-investing, and tax-loss harvesting constraints.

Key Insight: Final Summary: OptiVest AI shifts algorithmic finance from speculative return prediction to robust, constrained, and explainable portfolio decision-making. The system is fully realized across a production FastAPI backend, React dashboard, and rigorous quadratic optimization engine.